

Evaluating the Feasibility of Using Augmented Reality for Tooth Preparation

Takuya Kihara*
Johns Hopkins University
Tsurumi University

Andreas Keller
Johns Hopkins University
Technische Universität München

Takumi Ogawa
Tsurumi University

Mehran Armand
Johns Hopkins University

Alejandro Martin-Gomez†
Johns Hopkins University

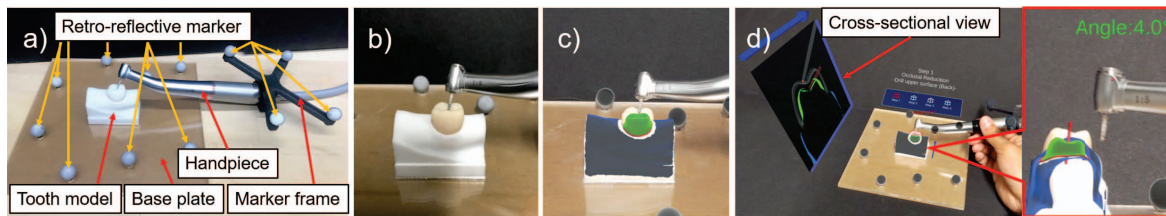


Figure 1: Setup and multiple scenarios of an experiment for tooth preparation using Augmented Reality (AR). a) Experimental setup. b) *Traditional*. Tooth preparation without AR. c) *Overlay*. Tooth preparation with AR system that tracks a tooth model to provide an ideal prepared tooth. d) *Cross-sectional*. Tooth preparation with AR system that tracks a tooth model and handpiece to provide a cross-sectional view of the anatomy and angle between a tool axis and references.

ABSTRACT

Tooth preparation is a fundamental treatment technique to restore oral function in prosthodontic dentistry. This technique is complicated as it requires the preparation of an abutment while simultaneously predicting the ideal shape. We explore the feasibility of using Augmented Reality (AR) Head-Mounted Displays (HMDs) to assist dentists during tooth preparation using two different visualization techniques. A user study (N=24) revealed that AR is effective for angle adjustment, and reduces the occurrence of over-reduction. These results suggest that AR can be used to assist physicians during these procedures and has the potential to enhance the accuracy and safety of prosthodontic treatment.

Index Terms: Computing methodologies—Computer graphics—Graphics systems and interfaces—Mixed / augmented reality; Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User studies

1 INTRODUCTION

Tooth preparation is a fundamental treatment technique of great importance in prosthodontics as it is essential for functional and durable prostheses. Deficient tooth preparation can be responsible for premature failures due to caries, periodontal disease complications, and detachment or fracture of teeth prostheses [2]. To prepare the tooth, the surgeons are required to select the shape (diameter and angle) of the dental bur as this will affect the reduction and taper during the preparation (Figure 3c). They need to consider the tooth axis, the clearance between the abutment tooth (Figure 3c) and the opposing tooth, and any potential teeth malalignment. Furthermore, because of the limited workspace, dentists often change their pose to adjust for the line of sight during the procedure. These factors complicate predicting the ideal abutment tooth shape and its preparation.

Preliminary work has explored the feasibility of using AR HMDs for tooth removal using Optical See-Through HMDs [3]. This work,

*e-mail: kihara-t@tsurumi-u.ac.jp

†e-mail: alejandro.martin@jhu.edu

introduced an AR system to assist users during root canal therapy in endodontic treatments and used optical markers and 3D printed models with 3.5 times larger size than the actual anatomy.

In this work, we introduce an AR system that provides navigation capabilities for tooth preparation in prosthodontics (Figure 1). Tracking algorithms that use the built-in cameras of AR HMDs to provide more accurate tracking of the tools used during these procedures [1]. To our knowledge, no other works focusing on this application is reported in the literature.

2 MATERIALS AND METHODS

2.1 Subject model

A mandibular model with a right first molar was scanned using a laser scanner magnified 1.5 times in size. The tooth models were fabricated using a 3D printer with bio-compatible resin. The base plate, fabricated using a laser cutter and a 3 mm acrylic sheet, had holes to fix retro-reflective markers to enable tracking.

2.2 System

The AR system used an Optical See-Through HMD (HoloLens2, Microsoft, WA, USA) to deliver the augmented content. To build and deploy an application, we used a workstation equipped with an Intel Xeon(R) W-11855M CPU, an NVIDIA RTX A5000 Laptop GPU, and 64 Gigabytes of Memory @3200 MHz. To track the objects of interest, we integrated algorithms capable of defining, detecting, and localizing the pose of passive retro-reflective markers (Figure 1a) [1]. The 3D-related components of the software were implemented using Unity 3D 2021.3.18f1 (Unity Technologies, NY, USA) and Visual Studio 2019 (Microsoft, WA, USA). Similar to the subject model, a rigid frame was fabricated to fix four retro-reflective markers that were attached to the dental handpiece (Figure 1a).

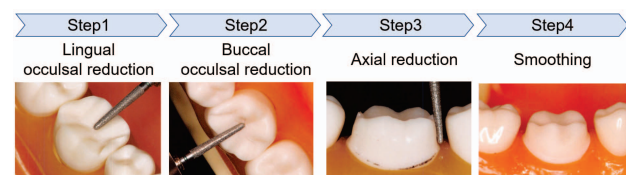


Figure 2: Multiple steps of tooth preparation procedure

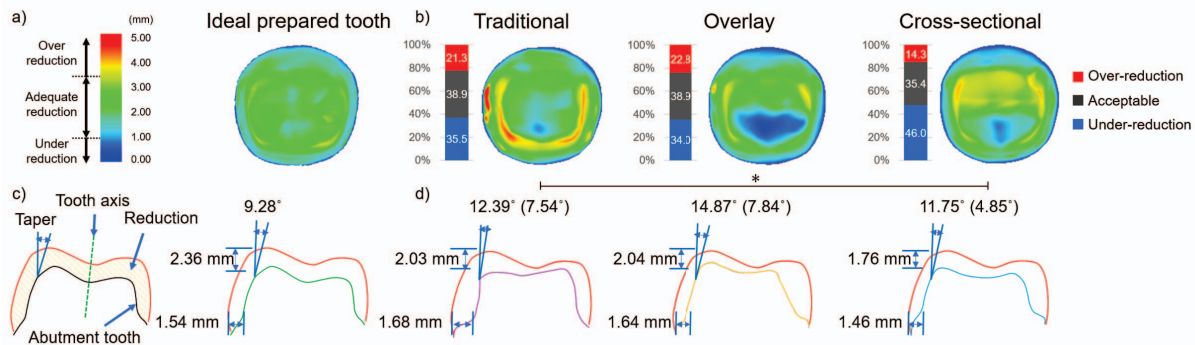


Figure 3: Results of the prepared teeth that participants shaved for each approach. a) The heat map shows the distance between the initial and final tooth shape. b) Percentages into three classes. c) Technical terms in dentistry. d) Buccal-lingual cross-section shows average distances between the initial (red) and final shape (orange, blue, and pink) at occlusal and axial surfaces, and the average angles and differences from the ideal angle (parentheses). The angle differences using *cross-sectional* were smaller and closer to the required angle than *traditional* (* $p < 0.05$).

The tooth preparation procedure consists of multiple steps. In this study, we defined a solution that included the following steps: 1) lingual occlusal reduction, 2) buccal occlusal reduction, 3) axial reduction, and 4) smoothing. The ideal abutment tooth and reference plane (or axis) were visualized through HoloLens2. The user was able to see the angle between the tool axis and the reference plane or pole to check if the bur angle was placed according to the planning data. The position of the tooltip was calculated by tracking, and buccolingual and mesiodistal cross-sectional views were displayed in real-time. The color of the bur would change from white to red when a collision with the ideal prepared tooth was detected, allowing the user to prevent excessive reduction.

2.3 Evaluation scenarios

We compared three different scenarios to evaluate the feasibility of fusing AR to assist surgeons during tooth preparation.

Traditional. Required the users to complete the task without the assistance of the AR HMD (Figure 1b).

Overlay. Allowed users to visualize the ideal prepared tooth superimposed over the anatomy using the AR HMD (Figure 1c).

Cross-sectional. Allowed users to visualize the ideal prepared tooth and cross-sectional views of the anatomy and tool (Figure 1d).

2.4 User study

To evaluate the feasibility of using AR for these dental procedures, we conducted a user study involving 24 volunteers (20 males and 4 females), whose ages ranged between 22 and 35 years (26.6 ± 3.7). All subjects gave written informed consent to participate in this study approved by the Johns Hopkins Institutional Review Board. None of the participants had experience with tooth preparation. Every participant prepared individual teeth using the three evaluation scenarios. To mitigate learning effects, the order of performing scenarios were randomly selected for each participant.

2.5 Variables

After task completion, the tooth models were laser scanned and their shape was compared to the ideal prepared tooth. We evaluated two morphological parameters as follows.

Distance to surface. This distance is calculated as the average of the point-to-point distances between the surfaces of the ideal shape and the prepared tooth.

Cross-sectional angles. These angles are calculated at the buccal-lingual cross-section between the tooth axis and the axial surface of the prepared tooth.

3 RESULTS AND DISCUSSION

To measure the distances between the ideal and prepared tooth surfaces, we obtained a heat map (Figure 3a) and percentages (Figure 3b) and classified the results into three classes: under-reduction ($d < -0.5\text{mm}$), acceptable ($-0.5\text{mm} < d < 0.5\text{mm}$), and over-reduction ($0.5\text{mm} < d$). The percentages for the *overlay* and *traditional* were similar (Figure 3b). However, the *cross-sectional* provided less over-reduction and more under-reduction than the other approaches. If tooth preparation shows under-reduction, the crown is thin and might cause its rupture. However, in such cases, dentists could perform additional shaving. On the other hand, since there is a nerve in the tooth, over-reduction might damage the nerve, which cannot be restored. These results suggest that *cross-sectional* approaches have the potential to support safer tooth preparation with less over-reduction. However, the system needs to be improved to make adequate reduction for fabrication of teeth prostheses.

The differences in angles between the ideal and actual prepared tooth on the buccal and lingual sides shaved using the *cross-sectional* approach (Figure 3d) were significantly smaller than the *traditional* ($p < 0.05$). There were no significant differences between *overlay* and the other approaches. The visualization of the cross-sectional views and the angular displays using *cross-sectional* for tooth preparation might also allow the user to properly angle the prepared tooth.

4 CONCLUSION

We proposed a system that explores the feasibility of using AR for tooth preparation. The visualization of *cross-sectional* approaches have the potential to support safer tooth preparation with less over-reduction than *traditional* and *overlay* approaches. The angular displays provided by the *cross-sectional* were considered helpful in tooth preparation, allowing the user to properly angle the prepared tooth. This system allows the user to confirm the morphology of the prepared tooth during the procedure for safer tooth preparation.

REFERENCES

- [1] A. Martin-Gomez, H. Li, T. Song, S. Yang, G. Wang, H. Ding, N. Navab, Z. Zhao, and M. Armand. STTAR: surgical tool tracking using off-the-shelf augmented reality head-mounted displays. *IEEE Transactions on Visualization and Computer Graphics*, 2023. doi: 10.1109/TVCG.2023.3238309
- [2] G. Ramesh, S. Nayar, and S.Chandrakala. Principles of Tooth Preparation - Review Article. *Indian Journal of Forensic Medicine & Toxicology*, 14:1340–1346, 2020. doi: 10.37506/ijfimt.v14i4.11715
- [3] T. Song, C. Yang, O. Dianat, and E. Azimi. Endodontic guided treatment using augmented reality on a head-mounted display system. *Healthcare Technology Letters*, 5:201–207, 2018. doi: 10.1049/hlt.2018.5062